

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 411 — Past Questions & Materials
by Renance

Course: MEE 411
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 400

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Paper 1 of 2 — MEE 411 Mechanical Engineering Design II

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING

First Semester Examination 2024/2025 Session

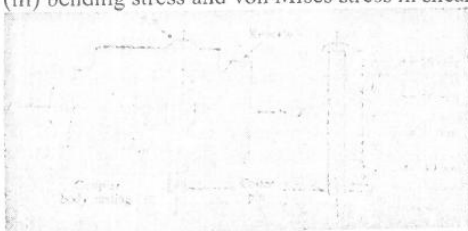
Course & Title : MEE 411 Mechanical Engineering Design II

Time Allowed: $2\frac{1}{2}$ Hours

Instruction: Answer five (5) questions.

(1a) Determine the loose running fit tolerances for a shaft and hole that have basic diameter of 40 mm. Given H11/c11 (14 marks)

2. For the loose pin in the figure below, determine (i) the effective beam span length, (ii) the largest bending moment, (iii) bending stress and von Mises stress in shear, and (iv) the ratio of von Mises stresses.



From the picture, $a = 56\text{mm}$, $b = 56\text{mm}$, $c = 94\text{mm}$, $d = 40\text{mm}$

(14 marks)

- 3a. Find the efficiency of a double riveted lap joint of 6mm plates with 20mm diameter rivets having a pitch of 65mm.
3b. Two lengths of mild steel tie rod having width 200mm and thickness 12.5mm are to be connected by means of a butt joint with double cover plates. If the permissible stresses are 80MPa in tension, 65MPa in shear and 160MPa in crushing. Calculate (i) the diameter of the rivet hole, (ii) Number of rivets (iii) thickness of the butt straps (iv) Make a sketch of the joint.

(14 marks)

- (4a) Derive an expression for a tensile strength of the joint for double fillet weld.
(4b) A plate 1m long, 50mm thick is welded to another plate at right angles to each other by 10mm fillet weld, find the maximum torque that the welded joint can sustain if the permissible shear stress intensity in the weld material is not to exceed 60MPa.

(14 marks)

- 5a. Calculate the circular pitch given the radius of the pitch circle to be 700mm and the number of teeth on the wheel are 40. Take $\pi = 22/7$.
5b A toothed wheel has 240 teeth, its module is 2mm. Find the pitch diameter, circular pitch and the diametral pitch.

(14 marks)

- 6b. Find the diameter of a solid shaft to transmit 15kW at 150rpm. The ultimate shear stress for steel is 300MPa and using a factor of safety of 5.

5c b. Why are stresses in a welded joint difficult to determine, state two assumptions to obtain such stresses.

(14 marks)

- 7a. A thin cylindrical pressure vessel of 1.2 m diameter generates steam at a pressure of 1.75 N/mm^2 . Find the minimum wall thickness, if (a) the longitudinal stress does not exceed 30 MPa, and (b) the circumferential stress does not exceed 40 MPa.

b. What are pressure vessels, explain the types of pressure vessels.

(14 marks)



$$p = \frac{2 \tau r}{\pi}$$

Paper 2 of 2 — MEE 411 mechanical engineering design 2024/2025

Level 400 • First Semester • PastQuestion

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DEPARTMENT OF MECHANICAL ENGINEERING

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(iii) bending stress and von Mises stress in shear, and (iv) the ratio of von Mises stresses.

From the picture, $a = 56\text{mm}$, $b = 56\text{mm}$, $c = 94\text{mm}$, $d = 40\text{mm}$

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3a. Find the efficiency of 2 double riveted lap joint of 6mm plates with 20mm diameter rivets having a pitch of 65mm.

3b. Two lengths of mild steel tie rod having width 20mm and thickness 12.5mm are to be connected by means of a butt

joint with double cover plates. If the permissible stresses are 80MPa in tension, 65MPa in shear and 160MPa in crushing.

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(14 marks)

6b. Find the diameter of a solid shaft to transmit 151 W at 15 Grpra. The ultimate shear stress for steel is 300MPa and

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